

SHOPMIND

The invisible decision brain for agentic commerce

Track 1: AI Shopping Agent | Kasparro Agentic Commerce Hackathon | April 2026

What this document is

A product thinking document — not a pitch deck. It documents what we are building, for whom, why, what we chose not to build, every tradeoff we made, and where this goes. The companion Technical Document covers architecture and implementation. The `DECISION_LOG.md` in the repo captures every live build decision.

1. The Problem

1.1 The shift that is already happening

In 2025, ChatGPT began completing purchases inside conversations. Google AI Mode started recommending products directly in search results. Shopify launched a free Agentic Plan connecting merchant catalogs to AI shopping channels. This is not a future trend — it is happening now.

And almost nobody has figured out how to make these AI shopping experiences actually good.

1.2 The real problem nobody has named

The browse-search-filter-compare loop — the dominant model of ecommerce for 25 years — was designed for a world where humans navigated information. It was never designed for a world where agents navigate on behalf of humans.

The result: AI shopping assistants that exist today are glorified search bars with a chat interface. They find products. They do not help you decide. They surface options. They do not narrow them. They show ratings. They do not address your specific hesitation. They hand you a list. They do not walk you to a purchase.

The gap we are filling:

Nobody owns the decision layer — the moment between 'I want something' and 'I bought it.' Search engines own discovery. Recommendation engines guess preferences. Payment gateways own the transaction. But the decision itself — understanding what the buyer actually needs, narrowing intelligently, building confidence, eliminating regret — is owned by nobody.

1.3 Why this problem is hard

The difficulty is not technical. It is psychological. Buyers — especially in gifting, personal care, and experience categories — do not know what they want. They know how they feel about what they want. They carry unstated constraints: budget anxiety they will not admit, relationship dynamics they cannot articulate, social signaling they are not aware of.

A keyword search cannot capture this. A filter panel cannot resolve it. A list of 47 products makes it worse. Only a conversation that asks the right questions, in the right order, without overwhelming the buyer, can navigate this.

2. Who We Are Building For

2.1 Primary user: the occasion-driven buyer

We are building for buyers who shop with emotional stakes — people purchasing for a reason, not just replenishing. The gifting category makes this visible and testable: a buyer shopping for their mother's birthday carries guilt, love, time pressure, and budget anxiety all at once.

This buyer type represents a significant portion of ecommerce purchases in categories like gifting, personal care, fashion, and home goods. According to RedSeer's India Ecommerce Report (2024), occasion-driven and gift-related purchases account for a substantial share of online retail in India, with festivals alone driving 35–40% of annual GMV for lifestyle categories. ShopMind is built for this emotionally complex buyer segment.

Buyer type	Current experience — what breaks
The anxious gifter	Spends 40 minutes browsing, buys the safest option, still unsure. Returns rate: 28% (Unicommerce Industry Report 2023).
The efficiency buyer	Filters aggressively, compares 6 similar products, cannot decide between them. Abandons cart.
The indulgent self-buyer	Knows vaguely what mood they want. Cannot translate mood into product. Exits to Instagram for inspiration.
The repeat buyer	Returns for same recipient's occasion. Has to start from scratch every time. Store has no memory.

2.2 Secondary user: the Shopify merchant

The merchant is not incidental — they are a co-user of ShopMind. Every conversation generates data they cannot get from any other source: which emotion keywords lead to purchase, where buyers hesitate, which products get high interest but low conversion, what categories have unmet demand.

Today, merchants guess at this from analytics dashboards that show clicks and exits. ShopMind gives them the reason behind every click and every exit.

Data privacy and DPDP compliance (L10):

The recipient shadow profile (L10) stores only the buyer's own stated preferences about a recipient — not the recipient's personal data. All profile data is session-scoped by default. Persistent cross-session storage requires explicit buyer opt-in and is governed by India's Digital Personal Data Protection Act (DPDP, 2023). The merchant dashboard (L11) receives only aggregated, anonymised signals — no individual buyer data is surfaced.

3. What We Are Building — V1 Scope

3.1 The one-line definition

ShopMind replaces the entire browse-search-filter-compare loop with a single conversation that understands what the buyer actually needs, narrows intelligently, recommends with confidence, and supports a clean path to purchase — without showing a single product list.

3.2 The core user journey

Every interaction follows the same spine, driven by the 11 integrated logics:

Stage	What ShopMind does	What the buyer experiences
1. Entry	Accepts any free-form input — text or mood tile selection	Zero friction start. No category to pick, no filter to set.
2. Intent parse (L1)	Extracts emotion vectors, occasion weight, relationship signal, unstated constraints	Feels heard. Not interrogated.
3. Narrowing (L2)	Asks exactly ONE smart follow-up question, cutting product space 60-70%	One question. Fast. Feels natural.
4. Comparison (L3)	If two products are close, compares on dimensions the buyer cares about	Understands the tradeoff. Makes an informed choice.
5. Recommendation (L4)	Delivers one product with a confidence score and plain-English reason	Knows why this was recommended. Trust is built.
6. Social proof (L7)	Shows persona-matched proof: who else bought this and in what context	Not just stars. Recognises themselves in the buyer.
7. Regret prevention (L8)	Pre-checkout: surfaces and answers the top objection proactively	Hesitation addressed before it forms. Proceeds with confidence.
8. Silent upsell (L9)	Offers one complementary product framed as completing the experience	Does not feel sold to. Genuinely completes the purchase.
9. Cart path (L5)	One-tap add to cart from within the conversation	No redirect. No product page hunt. Clean exit.

3.3 The mood canvas — non-verbal entry (L6)

Before typing a single word, the buyer can tap three mood tiles — words or textures like 'warm', 'minimal', 'bold', 'handmade'. ShopMind maps this aesthetic fingerprint to product emotion_tags in the catalog.

This matters because a significant fraction of buyers — especially in gifting — cannot articulate what they want verbally. They can recognise it visually or emotionally. The mood canvas is the entry point for this buyer type. No competing solution has non-verbal intent capture. This is a deliberate differentiation.

3.4 Technical foundation

ShopMind runs on a Shopify Partner development store, gifting category, approximately 40 synthetic products across skincare, accessories, home decor, and experience-adjacent items.

Component	Decision and reasoning
Shopify API	Admin API (GraphQL) — not Storefront API. Full catalog access, all metafields, variants, and inventory. Storefront API returns only published sales channel products with limited fields — insufficient for our metafield architecture.
Metafield schema	Three custom metafields per product: emotion_tags (e.g. warmth, nostalgia, luxury), persona (e.g. picky-mom, tech-dad, self-indulgent), occasion (e.g. birthday, anniversary, just-because). Foundation of L1 and L7.
AI layer	Claude via Anthropic API. System prompt enforces: one-question-at-a-time rule, JSON output with {product, confidence_score, reason, objection, complementary_product}, explicit AI/deterministic boundary.
Deterministic layer	Cart add, product fetch, metafield reads, confidence threshold logic, catalog validation — all deterministic. AI handles intent parsing and reasoning only. This boundary is documented in the Technical Document.
Category choice	Gifting. Chosen because: highest emotional complexity, most visible demonstration of L1 (emotional intent), most common occasion-driven purchase pattern, most measurable regret and return rate.

4. The 11 Integrated Logics

ShopMind is not a chatbot wrapper. It is a pipeline of 11 distinct logics, each satisfying a specific criterion from the Track 1 problem statement or judging dimension. Each logic has a clear input, output, and failure state documented in the Technical Document.

Logic	What it does	What it satisfies
L1 — Emotional intent parser	Extracts emotion vectors, occasion weight, relationship signal, unstated constraints from free-form text	Track 1 criterion 1: understand what the user actually needs
L2 — Progressive narrowing engine	One question per turn, cutting product space 60-70%. Psychographic decision tree, not keyword filter	Track 1 criterion 2: narrow options intelligently
L3 — Dimension-aware comparison	When confidence is split, compares products on buyer-stated dimensions — not specs	Track 1 criterion 3: compare products on dimensions that matter
L4 — Confidence scoring + reason	Every recommendation carries a numeric score and plain-English reason	Track 1 criterion 4: explain recommendations clearly

L5 — Frictionless cart path	One-tap add to cart from within the conversation. No redirect.	Track 1 criterion 5: clean path toward cart or checkout
L6 — Mood canvas (non-verbal)	Buyer taps 3 mood tiles. Aesthetic fingerprint maps to product emotion_tags. Zero typing required.	Originality 15%: goes beyond the obvious
L7 — Contextual social proof	Persona-matched proof from synthetic review metafields. Not aggregate stars.	Product Experience 20%: builds trust naturally
L8 — Regret prevention layer	Pre-checkout pass: surfaces and answers top objection proactively	Business Relevance 15%: directly reduces return rate
L9 — Silent upsell engine	One complementary product framed as completing the buyer's unstated goal	Business Relevance 15%: higher AOV without pushiness
L10 — Recipient shadow profile	Silent profile of gift recipient built from buyer answers (session-scoped, opt-in for persistence). DPDP-compliant.	Business Relevance 15%: LTV and repeat purchase engine
L11 — Merchant insight feed	Every conversation generates: which emotion tags convert, where buyers drop off, which products get high confidence but low conversion	Kasparro's core infrastructure mission: intelligence layer for commerce

5. What We Chose Not to Build — and Why

Scope decisions are as important as build decisions. The following were explicitly cut from V1 with documented reasoning.

Cut from V1	Reasoning
Product listing page	Showing a list defeats the purpose of the agent. If ShopMind outputs a list, it has failed. The entire value proposition is: one recommendation, not ten options. A listing page would be a regression to the old model inside the new one.
Search bar / filter panel	Adding these would let buyers bypass the conversation, reducing the quality signal for L1 and undermining the core flow. Explicitly not built.
Post-purchase story generator	Strong feature — generates a personalised gift message the buyer can send with the product. Cut because: validation of the core decision loop (L1-L5) is more important in V1. Documented as V2 with a clear implementation path.
Voice commerce layer	Voice input for tier-2/3 India buyers. Cut because the intent parsing prompt needs to be validated on text first. Voice adds transcription variability that should not be introduced before the core logic is stable.
Multi-category support	Gifting only in V1. A focused category produces better metafield coverage, cleaner training signal for L1, and a more coherent demo. Expanding categories is a configuration change, not a rebuild.
Real merchant store	Synthetic product data on a Shopify dev store per hackathon doc guidance. Real merchant data introduces catalog inconsistencies that

	would obscure whether failures are product or system failures.
WhatsApp / voice channels	ShopMind as a channel-agnostic intelligence layer is V3+. V1 validates the logic pipeline. Channel expansion is distribution, not product.
Multi-turn product returns in L9	Silent upsell limited to one offer. Showing multiple upsell options would re-introduce the list problem. One complementary product, framed correctly, is the constraint.

6. Key Decisions and Tradeoffs

6.1 One question per turn — the hardest constraint to hold

The most important product decision in ShopMind is the one-question-per-turn rule. Every instinct when building a recommendation system is to gather more information faster — ask about budget, occasion, recipient, and relationship all at once.

We chose not to. The reason: the buyer is not a database to be queried. Every additional question is a friction point. Research on conversational commerce shows that the single biggest cause of drop-off is the feeling of being interrogated. One question, asked well, builds trust. Four questions asked at once feels like a form.

The tradeoff: we need more turns to reach high confidence. We accept this. A buyer who completes three turns and buys is infinitely more valuable than a buyer who abandons after one turn with four unanswered questions.

6.2 Admin API over Storefront API

We use the Shopify Admin API, not the Storefront API. The Storefront API only returns products published to a specific sales channel and exposes limited fields — insufficient for our metafield architecture. Admin API gives full catalog access including custom metafields, which are the foundation of L1, L7, and L10.

The tradeoff: Admin API requires server-side calls for security. We accept this architectural overhead because the metafield data is irreplaceable — it is what makes ShopMind intelligent rather than just conversational.

6.3 Gifting category as the beachhead

Gifting is the hardest emotional context in commerce. If ShopMind can navigate a buyer who says 'something for my mom's birthday' to a confident purchase in three turns, it can handle any category. We chose the hardest problem to prove the system, not the easiest.

The tradeoff: gifting has lower average order value than electronics or furniture. We accept this because conversion rate and return rate impact are more important metrics for demonstrating business value at this stage.

6.4 Synthetic data with real metafield architecture

We use synthetic products but real metafield architecture. Every product has emotion_tags, persona, and occasion metafields that reflect genuine human buying patterns — drawn from research on actual gifting behavior, not invented arbitrarily.

The tradeoff: synthetic data cannot capture the long tail of real catalog messiness. We document this as a known limitation and describe the data quality process a production system would require.

6.5 Confidence score as a UI element

We expose the confidence score to the buyer. Most recommendation systems hide their uncertainty. We chose transparency: a buyer who sees 91% match understands why this product was recommended, and it creates a natural prompt for the agent to ask one more question when confidence is below 75%.

The tradeoff: showing uncertainty might reduce buyer confidence in the system. We mitigated this by framing the score as 'match strength' rather than system certainty — the language is about fit, not about the AI's limitation.

7. Failure Handling — What Happens When Things Break

The judging rubric explicitly evaluates failure handling — we treat it as a first-class product concern, not an afterthought. Full technical implementation is in the Technical Document.

Failure scenario	ShopMind's response
Shopify Admin API is down	Graceful message surfaces: 'Having trouble connecting to the store right now — try again in a moment.' No LLM call is made. No hallucinated product is presented. Cart functionality disabled until API recovers.
LLM returns malformed JSON	Deterministic retry once with a stricter prompt. If second response is also malformed: fallback to hardcoded clarifying question. Log the failure. Never present a broken recommendation.
LLM returns non-existent product_id	Deterministic validation checks every product_id against Admin API catalog before showing buyer. If invalid: re-prompt with explicit catalog context. Runs on every recommendation.
Buyer input is completely ambiguous	L1 parser returns low-signal flag. ShopMind responds with the mood canvas as fallback entry point — non-verbal input bypasses the ambiguity.
Confidence never exceeds 75% after 4 turns	ShopMind presents closest match transparently: 'This is the closest I found — here is why it fits and where it might not. Want me to look in a different direction?' Honest, not evasive.
Metafields missing on a product	Products with incomplete metafield coverage excluded from recommendation. Flagged for merchant review via L11. Ensures L1 always has complete signal to parse.
Network timeout mid-conversation	Conversation state is preserved in session. Buyer is shown a reconnecting indicator. On recovery, the last assistant turn is replayed from session state — no data loss.

8. Business Model and Commercial Relevance

8.1 How ShopMind moves merchant metrics

Merchant metric	How ShopMind moves it — with sourcing
Conversion rate	Replacing a 47-product list with one confident recommendation removes decision paralysis. Bloomreach's 2023 Personalization Report found conversational recommendation flows produced 18-32% higher conversion vs browse-and-filter UX for fashion and gifting categories.
Return rate	L8 (regret prevention) surfaces the top objection before checkout. Narvar's 2023 Consumer Returns Study found that 67% of returns are driven by buyer uncertainty at point of purchase — not product defect. Pre-empting that uncertainty directly reduces returns.
Average order value	L9 (silent upsell) adds one complementary product framed as experience completion. Dynamic Yield's personalisation benchmarks (2023) show contextual upsell in conversation flows drives 12-22% AOV lift vs static cross-sell carousels.
Repeat purchase rate	L10 (recipient shadow profile) makes the second purchase for the same recipient frictionless. Session-scoped by default; persistent with opt-in. Shopify's Commerce Trends 2024 report found that occasion-driven buyers who received a personalised second-purchase prompt returned at 2.4x the rate of un-prompted buyers.
Catalog intelligence	L11 surfaces unmet demand, pricing friction points, and high-interest/low-conversion products. Merchant acts on real signal from buyer psychology, not guesswork from click heatmaps.

8.2 Revenue model for ShopMind as a platform

V1 is a proof of concept on one store. The commercial model that emerges from it is usage-based SaaS: merchants pay per conversation resolved, not a flat subscription. This aligns ShopMind's incentives with merchant outcomes — we earn when buyers buy.

This model is directly aligned with Kasparro's stated mission: building the infrastructure for agentic commerce. ShopMind is the decision layer that sits above Shopify (the backend), above Google (the discovery layer), above payment gateways (the rails). The decision moment — owned by nobody today — is the most valuable layer in the stack.

9. What This Becomes — The Larger Vision

Note on scope:

The following roadmap documents the reasoning arc behind V1 decisions — not promises. Every V1 decision is made with this larger picture in mind.

9.1 Post-hackathon roadmap

Phase	What it is
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V1 — Hackathon (now)	11-logic gifting agent on one Shopify dev store. Validates: does the decision pipeline work? Does confidence scoring build buyer trust? Does regret prevention reduce hesitation?
Phase 1 — Automation (Month 1-3)	Autonomous inventory brain (demand sensing from conversations), dynamic pricing signals, post-purchase lifecycle engine, catalog self-improvement agent.
Phase 2 — Intelligence (Month 3-6)	Real-time buyer sentiment map, buyer persona auto-clustering from conversation data (not predefined segments), seasonal demand forecasting, catalog gap detector.
Phase 3 — Platform (Month 6-12)	Multi-merchant SaaS API (usage-based pricing), zero-touch merchant onboarding, cross-merchant anonymised intelligence network, white-label enterprise tier.
Phase 4 — Ecosystem (Year 2)	WhatsApp commerce agent (India-first, zero app download), voice commerce in regional languages, creator commerce integration, phygital retail QR agent.
Phase 5 — The moat (Year 2-3)	Commerce-native LLM fine-tuned on 10M+ ShopMind conversations. A model that understands commerce intent, buyer psychology, and objection handling better than any general LLM.

9.2 Advanced add-ons — real-world business extensions

The following are documented post-V1 extensions that transform ShopMind from a demo-grade shopping agent into a full-stack agentic commerce infrastructure layer. Each maps to a real business problem beyond the hackathon scope.

Add-on	What it does	Real business problem it solves
Autonomous inventory sensing	L11 signals aggregate across sessions into demand forecasting. When 40+ buyers in 7 days express interest in 'handmade wellness gifts under ₹800' but no product matches — the merchant gets a catalog gap alert with suggested sourcing brief.	Merchants stock by gut, not buyer intent. This makes stocking a data-driven process.
Dynamic pricing engine	Confidence score data + objection frequency feeds into real-time pricing signals. If 70% of buyers hesitate at ₹1,200 for a product but buy at ₹999, the system flags the price-resistance threshold.	Merchants set prices once. This makes pricing responsive to live buyer psychology.
Post-purchase lifecycle agent	After checkout, ShopMind triggers: a personalised gift message generator (V2 feature), a follow-up reminder 30 days before the next occasion, and a 'how did it go?' micro-survey that refines the recipient profile.	Post-purchase is a dead zone in most Shopify stores. This converts one purchase into a recurring relationship.
Cross-merchant intelligence network	Anonymised, aggregated signals across all ShopMind merchants surface macro trends: 'across 200 gifting stores, 'sustainable	Each merchant is an island. This builds the intelligence network that makes every merchant smarter.

	packaging' is the #1 objection for products above ₹1,500 in October.' No individual merchant can see this alone.	
WhatsApp commerce agent	ShopMind's entire 11-logic pipeline accessible via WhatsApp Business API. Buyer sends a voice note or text. ShopMind runs L1-L9 in conversation. Cart link delivered in-chat. Zero app download.	India's tier-2/3 ecommerce is WhatsApp-native. Most Shopify stores have zero presence there.
Creator commerce integration	A creator links their ShopMind store. Followers DM the creator's handle with a gift need. ShopMind runs the decision pipeline on the creator's curated catalog. Purchase attributed to the creator.	Creator commerce is growing 40% YoY in India. No tool makes creator-to-purchase frictionless without a full app.
Phygital QR retail agent	Retail stores embed QR codes on display products. Customer scans → ShopMind conversation opens → agent explains product, suggests variants, adds to cart for home delivery or reserve-at-counter.	Physical retail has zero intelligence layer. This brings the ShopMind decision engine into offline commerce.
B2B gifting workflow	Corporate buyers describe a gifting brief ('60 Diwali gifts, ₹800 each, for client retention'). ShopMind runs L1-L9 at bulk scale, generates a curated shortlist, handles bulk cart, and issues a pro-forma invoice.	Corporate gifting is a ₹12,000Cr market in India with no intelligent agent-layer solution.

Why these are not in V1:

Each add-on depends on the 11 core logics being validated first. V1 proves the decision pipeline works. These extensions prove it scales. Building them before the foundation is validated would produce a brittle system with impressive surface area and no depth.

9.3 The ceiling — beyond commerce

The 11 logics in ShopMind are not commerce-specific. Emotional intent parsing, progressive narrowing, confidence scoring, and regret prevention describe the structure of any high-stakes human decision under uncertainty.

Buying a gift for someone is structurally identical to choosing a hospital, a college course, a financial product, or an apartment. The same engine applies. Every major human decision domain has the same gap: discovery exists, transaction infrastructure exists, but the decision layer is owned by nobody.

ShopMind's long-term ceiling is not a shopping agent. It is decision infrastructure for any domain where humans must choose under emotional complexity and information overload. Commerce is the beachhead — the most frequent, most measurable, most commercially obvious entry point. But the problem being solved is much larger.

10. Alignment with Judging Criteria

Criterion (weight)	How ShopMind addresses it
Product Thinking & Documentation (25%)	This document. Decision log maintained from day 1 (DECISION_LOG.md in repo). Every scope cut documented with reasoning. What NOT built is as explicit as what IS built. Companion Technical Document covers architecture and failure handling.
Technical Execution & Architecture (25%)	Admin API (not Storefront). Metafield architecture. Deterministic/AI boundary explicit and documented. Failure handling documented for 7 failure scenarios in Technical Document. Atomic git commits. Architecture diagram in Technical Document.
Product Experience (20%)	Mood canvas (zero typing), one-question rule (no interrogation), confidence score UI (transparency), one-tap cart (no redirect), persona-matched social proof (recognition not ratings). Demo video in repo.
Business Relevance (15%)	Impact on 5 merchant metrics with sourced data (Bloomreach, Narvar, Dynamic Yield, Shopify benchmarks). Usage-based SaaS model documented. L11 directly serves Kasparro's infrastructure mission. DPDP compliance addressed.
Originality & Insight (15%)	Non-verbal mood input (no competitor has this), recipient shadow profile with privacy-first design, regret prevention before checkout, 8 post-V1 business extensions documented, merchant insight from conversation psychology.

What we are explicitly not:
A generic chatbot. A shallow LLM wrapper. A vague assistant with no product logic. A flashy demo with no practical value. Every logic maps to a specific Track 1 criterion. Every feature maps to a measurable merchant or buyer outcome. Every scope cut is documented with reasoning. Every business claim is sourced.

11. Team and Contribution

Role	Responsibilities
Product lead	Problem framing, user journey design, all documentation (Product Document, Technical Document, Decision Log), demo script, scope decisions, judging alignment
Engineering lead	Shopify Admin API integration, metafield architecture, Claude prompt engineering, confidence scoring pipeline, failure handling, git history, README
Shared	Mood canvas UI, demo video, metafield tagging of synthetic product catalog

11.1 AI tools used

Claude (Anthropic) used for: conversational agent core, intent parsing, confidence reasoning. Cursor used for code editing acceleration. All AI tool usage disclosed per hackathon eligibility rules.

All architectural decisions, product reasoning, and documentation are original work — AI tools were used to accelerate implementation, not replace thinking.

ShopMind is a focused, real solution to a real problem.

It starts with a 23-day build, a Shopify dev store, and 40 synthetic gifting products.

It ends with owning the decision layer of commerce.